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what is the 2D Fourier transform of 1/r^2?

In 2D, the Fourier transform of

$$f(\mathbf{r}) = \frac{1}{r^2}, \qquad r = \sqrt{x^2 + y^2}$$

is subtle because $1/r^2$ is not integrable at either $r = 0$ or $r \to \infty$. The transform must therefore be understood in the **distributional / regularized** sense.

Using the convention

$$\hat{f}(\mathbf{k}) = \int_{\mathbb{R}^2} d^2r \; e^{-i\mathbf{k}\cdot\mathbf{r}} f(\mathbf{r}),$$

the result is proportional to a logarithm:

$$\boxed{\mathcal{F}\left[\frac{1}{r^2}\right](\mathbf{k}) = -2\pi \ln|\mathbf{k}| + (\text{constant})}$$

More precisely, after introducing UV/IR cutoffs,

$$\int d^2r \; \frac{e^{-i\mathbf{k}\cdot\mathbf{r}}}{r^2} = -2\pi \ln\left(\frac{|\mathbf{k}|}{\Lambda}\right),$$

where Λ depends on the regularization scheme.

A quick derivation uses polar symmetry:

$$\hat{f}(k) = 2\pi \int_0^\infty \frac{dr}{r} J_0(kr),$$

and the standard Bessel integral identity

$$\int_0^\infty \frac{dr}{r} \left[J_0(kr) - 1\right] = -\ln k + \text{const.}$$

So in 2D:

- $1/r$ transforms to $1/k$,
- while $1/r^2$ transforms to $\ln k$.

The logarithm appears because $1/r^2$ is the **scale-invariant borderline case** in two dimensions.

For comparison, the related identity

$$\nabla^2 \ln r = 2\pi \delta^{(2)}(\mathbf{r})$$

implies

$$\mathcal{F}[\ln r] = -\frac{2\pi}{k^2},$$

which is often used to derive the logarithmic transform above.

tell me more about the standard bessel integral identity used

The key identity is

$$\int_0^\infty \frac{dr}{r} \left(J_0(kr) - 1\right) = -\ln k + C, \qquad k > 0,$$

where C is a k-independent constant that depends on how the divergent integral is regularized.

This identity is fundamental in 2D Fourier analysis because radial Fourier transforms reduce to Bessel integrals.

Why the subtraction $J_0(kr) - 1$?

The raw integral

$$\int_0^\infty \frac{dr}{r} J_0(kr)$$

diverges:

- near $r = 0$, since $J_0(0) = 1$, the integrand behaves like $1/r$,
- at large r, the oscillations are only conditionally convergent.

Subtracting 1 removes the short-distance divergence:

$$J_0(kr) - 1 \sim -\frac{k^2 r^2}{4} \quad (r \to 0),$$

so

$$\frac{J_0(kr) - 1}{r} \sim -\frac{k^2 r}{4},$$

which is integrable at the origin.

Scaling argument

The logarithm can almost be derived by dimensional analysis alone.

Define

$$I(k) = \int_0^\infty \frac{dr}{r}\,[J_0(kr) - 1].$$

Change variables:

$$u = kr, \qquad r = \frac{\nu}{k}, \qquad dr = \frac{d\nu}{k}.$$

Then

$$I(k) = \int_0^\infty \frac{d\nu}{\nu}[J_0(\nu) - 1].$$

At first glance this looks k-independent, but that hides the fact that the integral is only conditionally defined. Regularization introduces a scale, and comparing different k's gives

$$I(k) - I(k_0) = -\ln\frac{k}{k_0}.$$

Hence

$$I(k) = -\ln k + C.$$

This is characteristic of scale-invariant borderline integrals.

A cleaner derivation via differentiation

Define

$$I(k) = \int_0^\infty \frac{dr}{r}\,[J_0(kr) - 1].$$

Differentiate with respect to k:

$$\frac{dI}{dk} = \int_0^\infty \frac{dr}{r}\,\frac{d}{dk}J_0(kr).$$

Using

$$\frac{d}{dz}J_0(z) = -J_1(z),$$

we get

$$\frac{dI}{dk} = -\int_0^\infty dr\;J_1(kr).$$

Now substitute $u = kr$:

$$\frac{dI}{dk} = -\frac{1}{k}\int_0^\infty du\,J_1(u).$$

The standard integral

$$\int_0^\infty J_1(u)\,du = 1$$

gives

$$\frac{dI}{dk} = -\frac{1}{k}.$$

Integrating:

$$I(k) = -\ln k + C.$$

Connection to 2D Fourier transforms

For a radial function $f(r)$, the 2D Fourier transform reduces to a Hankel transform:

$$\hat{f}(k) = 2\pi\int_0^\infty r\,dr\,f(r)\,J_0(kr).$$

For

$$f(r) = \frac{1}{r^2},$$

this becomes

$$\hat{f}(k) = 2\pi \int_0^\infty \frac{dr}{r} J_0(kr),$$

which is exactly the divergent integral above. After regularization:

$$\hat{f}(k) \sim -2\pi \ln k.$$

Related classic identities

Some closely related Bessel integrals:

$$\int_0^\infty J_0(ax)\,dx = \frac{1}{a}, \qquad a > 0$$

$$\int_0^\infty x J_0(ax)\,dx$$

(diverges distributionally)

$$\int_0^\infty \frac{\cos(ax) - \cos(bx)}{x}\,dx = \ln\frac{b}{a},$$

which is essentially the ordinary Fourier analogue of the Bessel identity.

The Bessel version appears because circular symmetry replaces plane waves by $J_0(kr)$ in 2D radial problems.

what is the name of the 2nd order partial differential equation lap f = k^2 f

The equation

$$\nabla^2 f = k^2 f$$

is usually called the **Helmholtz equation** with a sign convention depending on context.

More specifically:

- The standard Helmholtz equation is often written as

$$\nabla^2 f + k^2 f = 0.$$

- Your version

$$\nabla^2 f - k^2 f = 0$$

  (equivalently $\nabla^2 f = k^2 f$)
  is commonly called the **modified Helmholtz equation** or sometimes the **screened Poisson equation** when sources are included.

The distinction matters because the solutions behave differently:

- Standard Helmholtz:

$$\nabla^2 f + k^2 f = 0$$

  gives oscillatory solutions like

$$e^{i\mathbf{k}\cdot\mathbf{r}}.$$

- Modified Helmholtz:

$$\nabla^2 f - k^2 f = 0$$

  gives exponentially growing/decaying solutions like

$$e^{-kr}, \quad e^{kr}.$$

In physics:

- Helmholtz appears in wave phenomena after separation of variables.
- Modified Helmholtz appears in diffusion with decay, screening, Yukawa potentials, Debye screening, etc.

In 2D radial coordinates, the modified Helmholtz equation becomes

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{df}{dr}\right) - k^2 f = 0,$$

whose solutions are the modified Bessel functions

$$I_0(kr), \qquad K_0(kr).$$

The decaying Green's function in 2D is proportional to

$$K_0(kr),$$

which behaves logarithmically near the origin:

$$K_0(kr) \sim -\ln r.$$

That logarithm is closely connected to the Fourier-transform behavior we discussed earlier.

So, by one regularization mechanism, the green's function of the 2D Laplace equation is ln(r) and, by another, it's a modified bessel function?

Yes — exactly. The two are closely related, and the modified Bessel function K_0 can be viewed as a regularized version of the logarithmic Green's function.

For the 2D Laplacian, the Green's function satisfies

$$\nabla^2 G(\mathbf{r}) = \delta^{(2)}(\mathbf{r}).$$

The fundamental solution is

$$G(r) = \frac{1}{2\pi} \ln r$$

(up to an additive constant and sign convention).

This logarithm is special to 2D.

Now consider the **modified Helmholtz operator**

$$(\nabla^2 - k^2)G_k(\mathbf{r}) = \delta^{(2)}(\mathbf{r}).$$

Its Green's function is

$$G_k(r) = \frac{1}{2\pi} K_0(kr),$$

where K_0 is the modified Bessel function of the second kind.

This acts as an infrared-regularized Green's function for the Laplacian.

Why K_0 approaches $\ln r$

The small-argument asymptotic is

$$K_0(z) \sim -\ln\left(\frac{z}{2}\right) - \gamma, \qquad z \to 0,$$

where γ is Euler's constant.

Substituting $z = kr$,

$$K_0(kr) \sim -\ln r - \ln\frac{k}{2} - \gamma.$$

So as $k \to 0$,

$$\frac{1}{2\pi}K_0(kr) = -\frac{1}{2\pi}\ln r + \text{constant depending on } k.$$

The divergent constant is physically irrelevant because Green's functions for the Laplacian are only defined up to additive constants.

Thus:

$$\boxed{\lim_{k\to 0}\left[K_0(kr) + \ln k\right] = -\ln r + \text{const}}$$

Fourier-space viewpoint

The relation becomes especially transparent in Fourier space.

For the Laplacian:

$$\tilde{G}(q) = -\frac{1}{q^2}.$$

This is singular at $q = 0$ (IR divergence).

The modified Helmholtz operator replaces it by

$$\tilde{G}_k(q) = -\frac{1}{q^2 + k^2}.$$

This is a regulated propagator.

Fourier transforming back gives

$$\frac{1}{q^2} \longleftrightarrow \ln r,$$

while

$$\frac{1}{q^2+k^2} \longleftrightarrow K_0(kr).$$

So K_0 is literally the massive/screened regularization of the logarithmic Green's function.

Physical interpretation

The parameter k introduces a length scale

$$\xi = \frac{1}{k}.$$

- Pure Laplace equation:
  long-range logarithmic behavior.
- Modified Helmholtz:
  exponential screening:

$$K_0(kr) \sim \sqrt{\frac{\pi}{2kr}} e^{-kr} \quad (r \to \infty).$$

So adding k^2 cuts off the long-range logarithmic growth.

This is exactly analogous to:

- Coulomb $\rightarrow$ Yukawa potential in 3D,
- massless $\rightarrow$ massive propagator in field theory,
- unscreened $\rightarrow$ screened electrostatics in plasmas.

is that the full asymptotical behavior of K_0(kr) as r -> infinity?

No. The expression

$$K_0(kr) \sim \sqrt{\frac{\pi}{2kr}} e^{-kr}$$

is only the **leading asymptotic term** as $r \to \infty$.

The full asymptotic expansion is an asymptotic series in inverse powers of kr:

$$K_\nu(z) \sim \sqrt{\frac{\pi}{2z}} e^{-z} \left[1 + \frac{4\nu^2 - 1}{8z} + \frac{(4\nu^2-1)(4\nu^2-9)}{2!(8z)^2} + \cdots \right].$$

For $\nu = 0$,

$$K_0(z) \sim \sqrt{\frac{\pi}{2z}} e^{-z} \left[1 - \frac{1}{8z} + \frac{9}{128z^2} - \frac{225}{3072z^3} + \cdots \right].$$

Substituting $z = kr$,

$$\boxed{K_0(kr) \sim \sqrt{\frac{\pi}{2kr}} e^{-kr} \left[1 - \frac{1}{8kr} + \frac{9}{128(kr)^2} - \cdots \right]} \qquad (r \to \infty).$$

The important qualitative features are:

1. **Exponential decay**

   $$e^{-kr}$$

   dominates.
2. **Algebraic prefactor**

   $$r^{-1/2}$$

   characteristic of 2D radial propagation.
3. **Asymptotic (not convergent) series**
   The power series generally diverges if summed to infinite order; it is meant to be truncated optimally.

You can derive this from the large-z saddle-point approximation to the integral representation

$$K_0(z) = \int_0^\infty e^{-z\cosh t}\, dt.$$

For large z, the dominant contribution comes from $t \approx 0$, where

$$\cosh t \approx 1 + \frac{t^2}{2},$$

giving

$$K_0(z) \approx e^{-z} \int_0^\infty e^{-zt^2/2}\, dt = \sqrt{\frac{\pi}{2z}} e^{-z}.$$

Higher-order corrections come from keeping higher powers in the expansion of $\cosh t$.